1: Title Slide

Title: Online Analytical Processing (OLAP)
Subtitle: Deep Dive into Analytical Systems

📌 Slide 2: What is OLAP?

Definition:
OLAP stands for **Online Analytical Processing**. It enables users to analyze multidimensional data interactively from multiple perspectives, supporting decision-making processes.

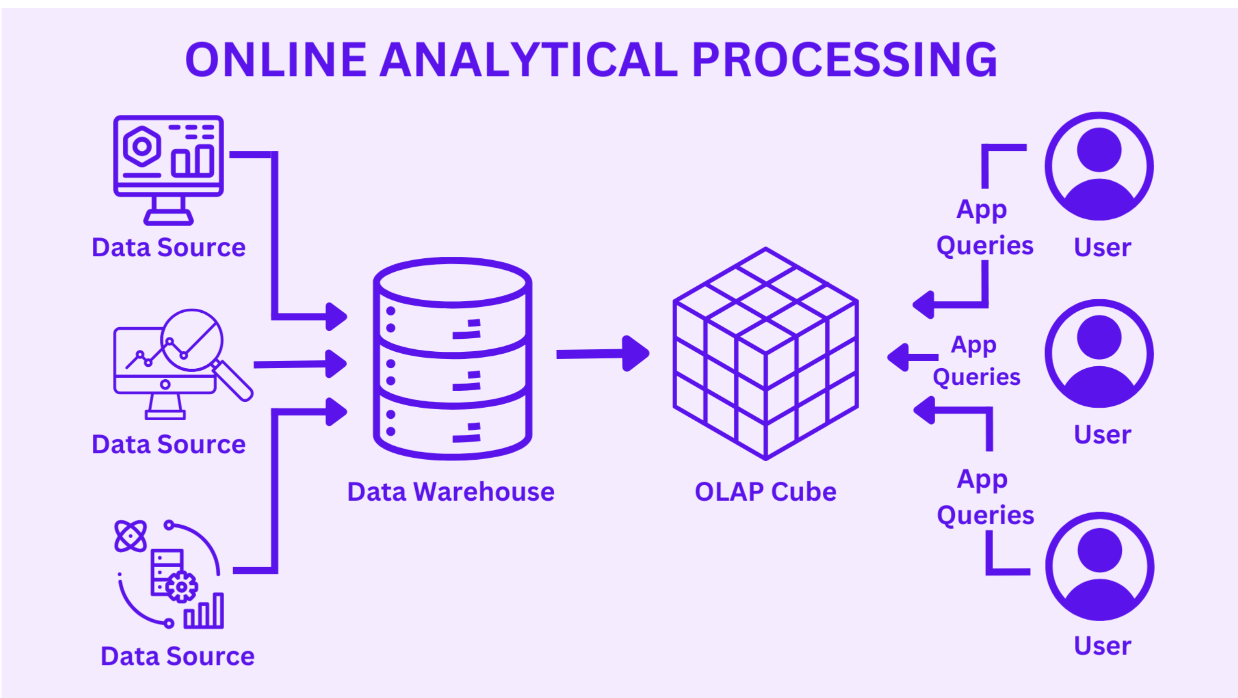

Illustration:

- A cube showing dimensions: Product, Time, Region.

📊 Slide 3: Nature of OLAP Analysis

Characteristics:

- **Multidimensional views** of data (e.g., slicing, dicing, drilling).
- **Aggregated computations** (SUM, AVG, COUNT).
- **Interactive analysis** (Pivot, filter).
- **Ad-hoc query capabilities**.

Code Sample (SQL Equivalent of Slice):

```
SELECT region, SUM(sales)
FROM sales_data
WHERE product = 'Laptop'
GROUP BY region;
```

🧱 Slide 4: Types of OLAP

Type	Description	Example
MOLAP	Multidimensional OLAP – uses multidimensional cubes.	Microsoft SSAS
ROLAP	Relational OLAP – uses relational databases.	Oracle BI
HOLAP	Hybrid OLAP – combination of MOLAP and ROLAP.	SAP BW

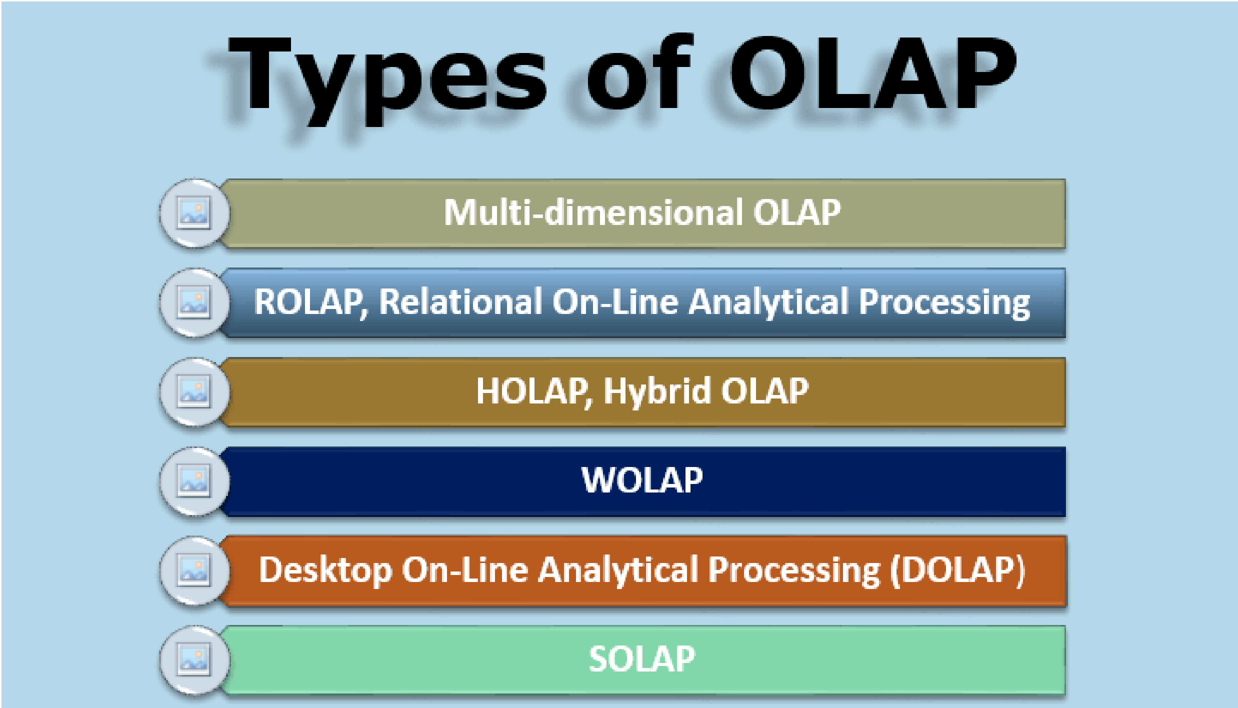

Illustration: Venn diagram showing overlaps of ROLAP, MOLAP, and HOLAP.d

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🛠️ Slide 5: OLAP Tools

- **Microsoft SSAS**
- **SAP BW**
- **Pentaho**
- **IBM Cognos**
- **Oracle OLAP**

Features to Highlight:

- Drill-down/up
- Pivot
- Dashboard integration

🔄 Slide 6: OLTP vs OLAP

Feature	OLTP	OLAP
Purpose	Day-to-day transaction	Decision support & analysis

Data Volume	Low per transaction	Huge historical data
Operations	Insert, Update, Delete	Select (complex queries)
Data Structure	Normalized	Denormalized (star/snowflake)
Example	Banking system	Sales forecasting tool

✅ Slide 7: OLAP Functional Requirements

- Support **multidimensional analysis**
- Ability to **drill-down / roll-up**
- Fast **aggregation** operations
- **Data security** per role/user
- High **query performance**
- **Time Intelligence** support

⚡ Slide 8: Why OLAP is Fast and Selective?

Reasons:

- Pre-aggregation of data.
- Indexing on dimensions.
- Storage optimization using **data cubes**.
- **Caching** of previous queries.

Illustration: Query Flowchart: Query → Cube Cache → Aggregated Result

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🔍 Slide 9: Operational vs Informational System

Feature	Operational System	Informational System
Users	Frontline workers	Analysts, Managers
Data	Real-time	Historical
Focus	Transactions	Insights

Examples	CRM, ERP	Data Warehouse, OLAP Tools

🧠 Slide 10: Additional Section – Use Cases of OLAP

- Sales forecasting by region/product
- Marketing campaign analysis
- Financial budgeting and variance
- Inventory performance review

📊 Slide 11: Illustration – OLAP Cube Example

Cube Axes:

- **X-axis**: Time
- **Y-axis**: Region
- **Z-axis**: Product
  Measure: Total Sales

Excel Dashboard Idea:

- Create a pivot table using OLAP data source with slicers for Year, Product, Region.

🔚 Slide 12: Summary

- OLAP is crucial for multidimensional data analysis.
- Multiple OLAP types suit different infrastructures.
- Integrates with BI tools to derive **actionable insights**.
- OLTP ≠ OLAP – they serve different goals.